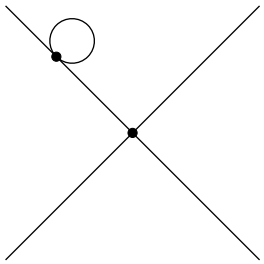
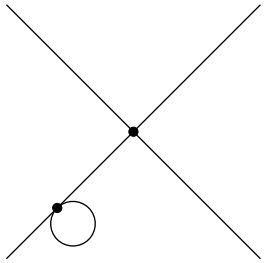
Contact-leg seagulls: four placements

Two quartic interaction vertices and one self-loop: six edges. Plain lines; no momentum, caps or circles.

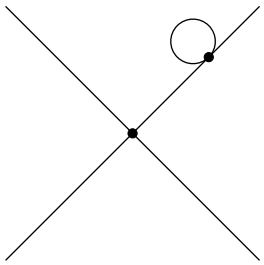
Slot 1: upper left



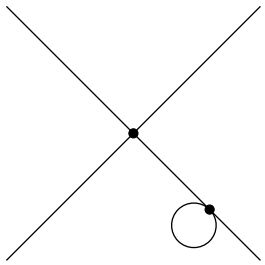
Slot 2: lower left



Slot 3: upper right



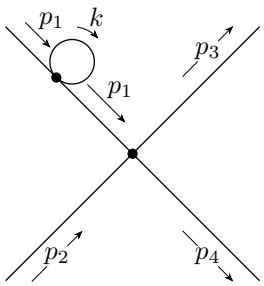
Slot 4: lower right



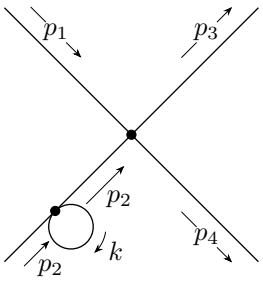
Momentum labels and loop flow

Defaults: p1/p2 in, p3/p4 out. Loop labels clear their local curve normals.

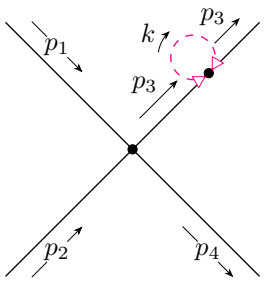
Slot 1: default flow



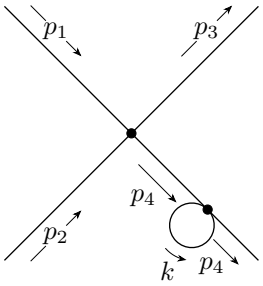
Slot 2: default flow



Slot 3: pink loop



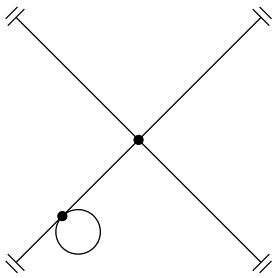
Slot 4: reversed loop



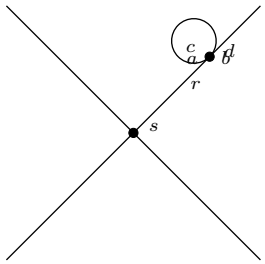
Separate controls and compatibility

Caps and indices appear only in their dedicated checks. Index appearance is deferred (D01).

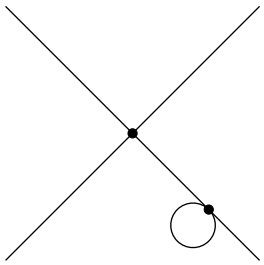
Slot 2: outer caps only



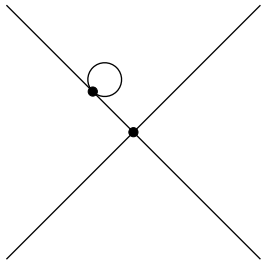
Slot 3: independent indices



Slot 4: explicit top orientation



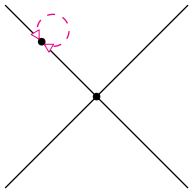
Slot 1: shifted and smaller



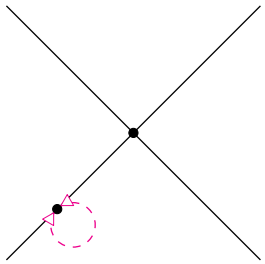
Proca endpoint contact: all four legs

Only the self-loop is pink; two open markers meet the same quartic insertion. No caps, circles or momentum.

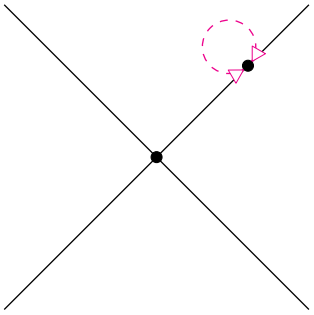
Slot 1; scale .72



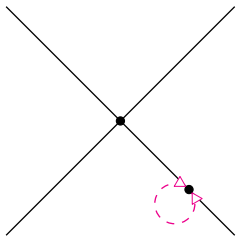
Slot 2; scale 1



Slot 3; scale 1.2



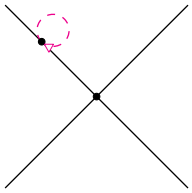
Slot 4; scale .9



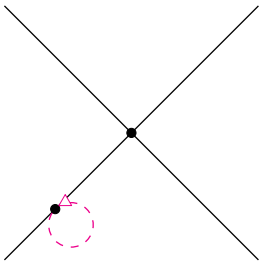
One-sided scalar endpoint contact

One open marker per loop. No caps, circles or momentum.

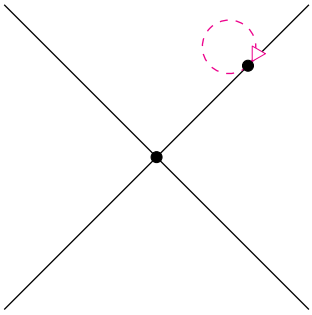
Slot 1: spolin; scale .72



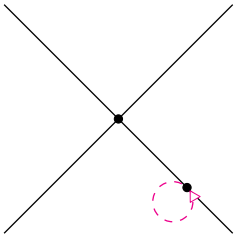
Slot 2: spolout; scale 1



Slot 3: spolin; scale 1.2



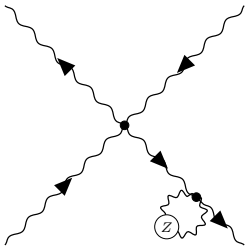
Slot 4: spolout; scale .9



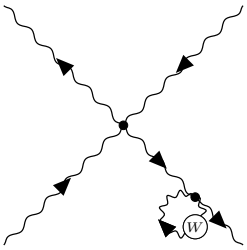
Slide 11 species and charge-flow variants

Eight source panels: neutral Z or directed W loop at each corner. Filled charge-flow arrows are separate from momentum guides (hidden here). No outer caps.

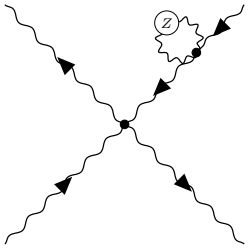
S11-T1: lower right Z



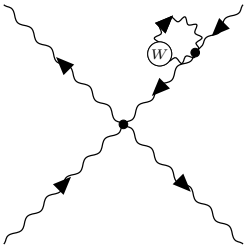
S11-T2: lower right W



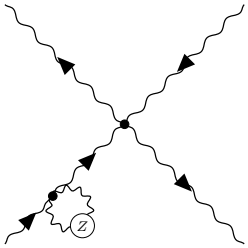
S11-T3: upper right Z



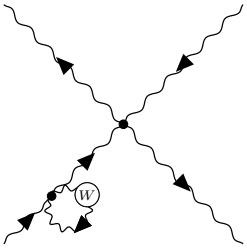
S11-T4: upper right W



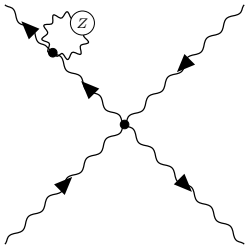
S11-B1: lower left Z



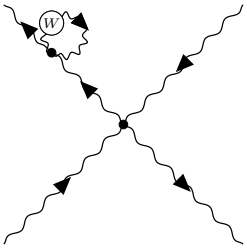
S11-B2: lower left W



S11-B4: upper left Z



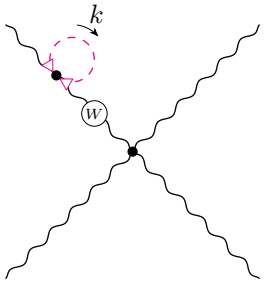
S11-B3: upper left W



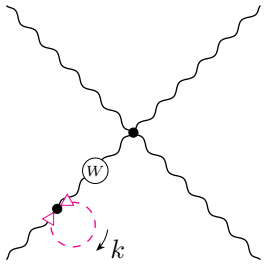
Styled samples: pink scalar polarization and inner W

EW outer legs without caps/circles; inner W circle; pink loop with two open endpoint markers, no circle and momentum k.

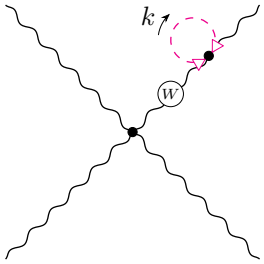
Slot 1



Slot 2



Slot 3



Slot 4

